

ELEC4631 Quiz 2

Instructions:

- This is an assessment item. You may not discuss the quiz problems or post your solutions on the Moodle student's forum. Violating this will be considered as academic misconduct.
- **Submitted solutions must be your own work**, not gained through plagiarism (copying someone's else work). Plagiarism is an academic misconduct and can result in zero marks for both the plagiator and the source.
- **All analytical workings must be shown in your solution. No workings and demonstration of the solution steps, no marks. Solutions by Matlab only will receive no marks.**
- You will submit your assignment via Moodle. Only submissions in pdf format will be accepted. Your assignment may be typed up or handwritten and scanned. Handwritten and scanned solutions must be clear and readable, and scans should not be in very high resolution that results in a large file size. Unreadable submissions will not be marked.
- Due by date: Wednesday 19/07/2023, 5 pm. Penalty for late submissions apply per the course outline.

Total marks for this quiz: 100

1. **(40 marks in total)** Consider the linearised quantum circuit from Quiz 1 Problem 4 with the state-space equation:

$$\begin{aligned}\dot{x} &= \begin{bmatrix} -\frac{1}{C}(2 + 1/R) & -1/C \\ 1/L & 0 \end{bmatrix} x + \begin{bmatrix} \frac{1}{RC} \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} x.\end{aligned}$$

Let $C = 0.5$ F, $L = 0.2$ H and $R = 2$ ohms.

- (a) (30 marks) Using the Ackermann's formula twice, design a linear output feedback control law with gain K and a Luenberger observer with gain L so that the closed-loop system (consisting of system + output feedback + observer) has the characteristic polynomial $c_{A_{cl}}(s) = (s^2 + 6s + 13)(s + 2)(s + 3)$, with the observer contributing only real eigenvalues (recall that A_{cl} denotes the closed-loop A matrix).

(b) (10 marks) Sketch a block diagram of the closed-loop system.

2. **(40 marks in total)** For the same circuit as in Problem 1, consider the infinite horizon LQR cost

$$J_{\infty}(x_0, u) = \int_0^{\infty} ((x_1(t) - x_2(t))^2 + 4u(t)^2) dt.$$

Answer the following questions:

- (a) (10 marks) Verify that the infinite horizon problem has a solution.
- (b) (30 marks) Compute the unique solution of the CARE equation satisfying $P > 0$ and determine the optimal control law u_{opt} .

3. **(20 marks total)** Consider the system

$$\dot{x} = \begin{bmatrix} \delta - 1 & 1 \\ -1 & \delta + 1 \end{bmatrix} x + \begin{bmatrix} \epsilon \\ \epsilon \end{bmatrix} u,$$

where $x = (x_1, x_2)^T$, and δ and ϵ are unknown real parameters. Answer the following questions:

- (a) (4 marks) Show that the system is not controllable for any value of the parameter ϵ .
- (b) (7 marks) Show that there is a linear combination of x_1 and x_2 that evolves independently of the input u .
- (c) (9 marks) Using the result of sub-problem (b) and **WITHOUT** computing the characteristic polynomial of $A - BK$ (you may, however, compute the characteristic polynomial of A), show that there is an eigenvalue of A that cannot be moved by any linear state feedback law $u = -Kx$ (that is, for any choice of K , $A - BK$ will always have this eigenvalue).